
Problem 9.13. Consider the function *pad*: $\Sigma \times \mathbb{N} \longrightarrow \Sigma^*\#^*$ that is defined as follows. Let $pad(s, l) = s\#^j$, where $j = \max(0, l - m)$ and m is the length of s. Thus, $pad(s, l)$ simply adds enough copies of the new symbol # to the end of s so that the length of the result is at least l. For any language A and function f: $\mathbb{N} \longrightarrow \mathbb{N}$, define the language $pad(A, f)$ as

$$pad(A, f) = \{pad(s, f(m)) \mid \text{where } s \in A \text{ and } m \text{ is the length of } s\}.$$

Prove that if $A \in \text{TIME}(n^6)$, then $pad(A, n^2) \in \text{TIME}(n^3)$.

Solution. □